Grade 8 Name_Answer Key________
Unit 14 Date____________________
Lesson 5 Practice Period__________________

The local faire had a game where a player chooses a rubber duck from a small tub of water, replaces it, and then chooses another rubber duck. There are a total of ten rubber ducks, all the same size and color, but two of the rubber ducks have a black dot drawn on the bottom. If a player chooses a duck with a black dot on the bottom, they win a prize.

1. Calculate the theoretical probability that a player will choose a rubber duck with a dot and a rubber duck without a dot.

   $\mathrm{P}(\text{dot}) = \frac{2}{10} = \frac{1}{5}$ $\mathrm{P}(\text{no dot}) = \frac{8}{10} = \frac{4}{5}$

   $\mathrm{P}(\text{dot, no dot}) = \frac{1}{5} \cdot \frac{4}{5}$
   $= \frac{4}{25}$

2. Calculate the theoretical probability that a player will not choose a duck with dot at all.

   $\mathrm{P}(\text{no dot at all}) = \frac{4}{5} \cdot \frac{4}{5} = \frac{16}{25}$

3. Calculate the theoretical probability that a player will choose a duck with a dot both times.

   $\mathrm{P}(\text{dot both times}) = \frac{1}{5} \cdot \frac{1}{5} = \frac{1}{25}$

A chip with one red side and one white side is tossed 14 times.

4. What is the theoretical probability that the chip lands on red half the time and white the other half?

   $$\left(\frac{1}{2}\right)^{14} = \frac{1}{16{,}384}$$

5. What is the theoretical probability that the chip lands on white all 14 times?

   $$\frac{1}{16{,}384}$$

Dante and his brother are arguing over taking out the trash and doing the dishes. They decide to roll a fair six-sided die twice to decide who has to do which chore. If the die lands on an even number, Dante will have to do that chore. If die lands on an odd number, Dante's brother will have to do that chore.

6. What is the theoretical probability that Dante will have to take out the trash and do the dishes?

$\text{P(even)} = \frac{3}{6} = \frac{1}{2}$

$\text{P(even, even)} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

7. What is the theoretical probability that Dante will *not* have to do either chore?

$\text{P(odd)} = \frac{3}{6} = \frac{1}{2}$

$\text{P(odd, odd)} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

Tonia has to read two different books for her class this grading period. She can't decide on what genres to read. Her teacher suggests using the spinner shown.

8. Calculate the theoretical probability that she will read a graphic novel and a historical fiction book.

$$\frac{1}{5} \cdot \frac{1}{5} = \frac{1}{25}$$

9. Calculate the theoretical probability that she will read two science fiction books.

$$\frac{1}{5} \cdot \frac{1}{5} = \frac{1}{25}$$

10. Calculate the theoretical probability that she will not read any nonfiction at all.

$$\frac{4}{5} \cdot \frac{4}{5} = \frac{16}{25}$$